

Designation: D 5678-24

Standard Specification for Aramid Fiber Ballistic Protection Fabric

This standard was issued on April 12, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers high-strength aramid fiber woven fabrics for ballistic protection applications

1.2 The aramid ballistic fabric covered by this specification are intended for use in manufacturing of soft body armor, helmets, vehicle armor panels, and protective equipment for law enforcement and military personnel.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

D 7269 Test Methods for Tensile Testing of Aramid Yarns

D 3822 Test Method for Tensile Properties of Single Textile Fibers

D 3776 Test Methods for Mass Per Unit Area (Weight) of Fabric

NIJ 0101.06 Ballistic Resistance of Body Armor

3. Terminology

3.1 Definitions:

3.1.1 aramid fiber—a class of heat-resistant and strong synthetic fibers with aromatic polyamide structure

3.1.2 denier—unit of linear mass density of fibers equal to mass in grams per 9000 meters

3.1.3 ballistic performance—the ability of material to resist penetration by projectiles

4. Classification

4.1 aramid ballistic fabric shall be classified according to fiber type, fabric construction, and areal density.

4.1.1 **Style 328**—Para-aramid, plain weave, 200 g/m², threat level IIA

4.1.2 **Style 706**—Para-aramid, basket weave, 440 g/m², threat level IIIA

4.1.3 **Style 929**—Meta-para aramid blend, plain weave, 360 g/m², threat level II

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Style (328, 706, or 929)

5.1.3 Width (cm)

5.1.4 Quantity (linear meters or m²)

5.1.5 Color (natural yellow, or custom dyed)

5.1.6 Edge finish (selvage or cut)

6. Materials and Manufacture

6.1 Yarns shall be made from 100% virgin aramid fiber with no reprocessed content.

6.2 Para-aramid fibers shall have minimum tensile strength of 3.0 GPa.

6.3 Weaving shall be performed on modern looms with appropriate tension control.

7. Mechanical and Physical Requirements

Property	Requirement	Test Method
Tensile strength (warp)	≥ 2200 N/cm	ASTM D 5034
Tensile strength (fill)	≥ 2200 N/cm	ASTM D 5034
Areal density	$\pm 5\%$ of nominal	ASTM D 3776
Thickness	0.25-0.75 mm	ASTM D 1777

8. Dimensions and Permissible Variations

8.1 Standard fabric width is 150 cm $\pm$ 2 cm.

8.2 Fabric shall be supplied in continuous rolls of specified length.

8.3 Selvage width shall not exceed 15 mm on each side.

8.4 Bow and skew shall not exceed 2% of fabric width.

9. Workmanship, Finish, and Appearance

9.1 Fabric shall be free from holes, tears, broken yarns, oil spots, and other defects that would impair ballistic performance. Weave shall be uniform and tight.

10. Sampling

10.1 Sample testing shall be conducted per MIL-STD-105 with AQL of 1.5. Each production lot shall undergo verification testing.

11. Test Methods

11.1 **Tensile Testing**—Cut specimens in warp and fill directions and test per ASTM D 5034 grab test method.

11.2 **Ballistic Testing**—Construct test panels per NIJ 0101.06 and conduct ballistic testing at accredited facility.

11.3 **Fabric Analysis**—Determine thread count, crimp, and fabric construction parameters using microscopy and mechanical analysis.

12. Inspection

12.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

13. Rejection and Rehearing

13.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

14. Certification

14.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

15. Product Marking

15.1 Fabric rolls shall be labeled with style number, lot number, width, length, mass per unit area, and production date.

16. Packaging and Package Marking

16.1 Fabric shall be rolled on cores and wrapped in protective polyethylene. Rolls shall be stored in cool, dry conditions away from UV exposure.

17. Keywords

17.1 aramid; ballistic; body armor; Kevlar; protective fabric; bulletproof; high strength